Chapter 1: Distributed Consensus Protocols

1.1 Two-Phase Commit Protocol (2PC)

The Two-Phase Commit Protocol is an atomic commitment protocol for distributed databases.

Phase 1 (Prepare Phase): The coordinator node sends a PREPARE message to all cohort participants.

Phase 2 (Commit Phase): If all cohorts reply YES, the coordinator sends a GLOBAL-COMMIT command. If any cohort replies NO or times out, the coordinator sends a GLOBAL-ABORT command.

A key limitation of Two-Phase Commit Protocol is that it is a blocking protocol if the coordinator crashes during the commit phase.

Chapter 2: State Machine Replication

2.1 Raft Leader Election and Log Replication

Raft is a consensus algorithm that manages a replicated log.

Raft achieves consensus by electing a distinguished leader node. The leader accepts log entries from clients and replicates them across followers.

Raft uses randomized election timers ranging between 150ms and 300ms to prevent split-vote scenarios.